Indian Institute of Technology Jodhpur

No. IITJ/OA/C/2020-21/19/2645 **Dated: 31 July 2021**

CIRCULAR

Subject: Engineering science options for B.Tech. students joined post July 2020

I. Engineering science options

To provide flexibility in curricula, options have been created to pursue

1. **4-year B.Tech. in Engineering Science**: Any student may opt for B.Tech. in Engineering science post 1st year to pursue a targeted focus area in Science and Engineering to meet demands of emerging industrial and academic requirements. Students will be required to complete core course requirements in the first two years of study whereas they can outline their own curriculum in the form of electives for the remaining two years from courses across disciplines of engineering, science, and humanities. The students will be awarded B.Tech in Engineering Science. Similar to regular B.Tech. Programme Students can also use *10 credits* out of stipulated electives and 10 additional credits to earn minor/specialization as an add-on to the B.Tech degree.
2. **4-year B.Tech. in X + Engineering Science**
   Students can pursue a capability linked B.Tech. in X + Engineering Science to pursue newer structures of programmes in addition to classical programmes, where 'X' represents the primary branch in which a student has taken admissions. After completion of the first year, the student having a CGPA of 6.5 or more may opt to choose Capability-linked B.Tech. in X + Engineering Science. Students will be required to register for extra credits for the same. The students pursuing this programme will take common courses and courses with a concentration in Engineering Science in the 2nd year. From the third year onwards the student would also start opting for courses related to the primary branch (X) and elective courses in the area of both primary branch and Engineering Science. At the end of the second year, an option of primary branch change will also be available to meritorious students of this programme. The students will be awarded B.Tech. in X and Engineering Science, e.g., B.Tech. in Mechanical Engineering and Engineering Science. Students of this programme would not be able to opt for any other specialization/minor.
3. **3-year B.Sc. in Engineering Science**:
   This option is designed mainly for underperforming students as an exit plan**.** The student would be required to complete required credits under institute and engineering science core, and open electives categories.

Course Categories and credit distribution of the proposed Engineering Science Programmes are given in Table 1.

Table 1. Course Categories and credit distribution of Engineering Science Programmes

	Course Type	Course Category	B.Tech. Regular		B.Tech. ES		B.Tech. in X + ES		B.Sc. ES	
			Cr	To	Cr	To	Cr	To	Cr	To
1	Institute Core (I)	Engineering (IE)	34	**69**	34	**59**	34	**59**	34	**59**
		Science (IS)	16		16		16		16	
		Humanities (IH)	12		9		9		9	
2	Programme Linked (L)	Science (LS)	7		0		0		0	
3	Programme Core (P)	Programme Compulsory (PC)	50	**71**	-	**-**	50	**71**	-	-
		Programme Electives (PE)	18				18			

		B.Tech. Project (PP)	3				3			
4	Open (O)	Open Electives (OE)	10	**10**	61	**61**	-	**-**	19	**19**
5	Engineering Science (E)	Engineering Science Core	0	**0**	22	**30**	22	**30**	22	**22**
		Engineering Science Elective (EE)	0		8		8		0	
		Total Graded		**150**		**150**		**160**		**100**
6	Non-Graded (N)	Humanities (NH)	6	**15**	6	**15**	6	**15**	5	**10**
		Engineering (NE)	3		3		3		3	
		Design/Practical Experience (ND)	6		6		6		2	
		Total Graded + Non-Graded		**165**		**165**		**175**		**110**

II. Engineering Science Compulsory and Elective Courses

- **Engineering Science compulsory courses**
    1. Probability, Statistics and Stochastic Processes (3-1-0)
    2. Modern Physics (3-0-0)
    3. Embedded Systems and IoT (3-0-2)
    4. Data Structures and Algorithms (3-0-2)
    5. Design and Analysis of Experiments (3-0-0)
    6. Modelling and Simulation (3-0-2)
- **Engineering Science Elective courses**

Courses with Science Concentration

1. Scientific Computations
2. Foundations of Quantum Information Processing
3. Physics of Complex Liquid
4. Analytical Mechanics
5. Nanoscience
6. Electronic Materials
7. Magnetic Materials
8. Energy Materials
9. Optical Measurement Techniques

Courses with Interdisciplinary Concentration

10. Principles of Biomechanics
11. MEMS and Microsystems
12. Non-Conventional Sources of Energy and Emerging Technologies
13. Modern Control
14. Sensors and Actuators
15. Measurement and Instrumentation
16. Digital Image processing
17. Computer Vision
18. Autonomous Systems
19. Introduction to Industry 4.0
20. Foundations of Cyber-Physical Systems
21. Smart Manufacturing
22. Introduction to Robotics
23. Introduction to Data Science

Note: In the case of a course being the common core of the primary branch and Engineering science, the course would be double-counted towards the core requirement of both. However, students will be required to take extra credit equivalent to total credits of all courses with double counting under the Engineering Science elective category. For example, students of Mechanical engineering doing double B.Tech. have Data Structures and algorithms (4 credits) as a common core course for both mechanical engineering and Engineering science degrees. Hence, this course will be double-counted towards the core course requirement for the award of the degree. In this case, students will be required to take an extra 4 credits worth of engineering science electives.

II. Programme Structure of Engineering Science Programmes

Table 2. Programme structure of 1st and 2nd year of B.Tech. in 1) Engineering Science and 2) X + Engineering Science

Cat	Course	LTP	CH	NC	GC	Cat	Course	LTP	CH	NC	GC
	I Semester						**II Semester**				
As per the curriculum of Primary Branch X [33 Graded Credit and 6 Non-graded Credits]											
	III Semester						**IV Semester**				
IE	Computational Material Design	1-0-0	1	-	1	IE	Materials Selection	1-0-0	1	-	1
ES	Probability, Statistics and Stochastic Processes	3-1-0	4	-	4	IE	Structure of Materials	1-0-0	1	-	1
ES	Modern Physics	3-0-0	3	-	3	ES	Embedded Systems and IoT	3-0-2	5	-	4
IE	Thermodynamics	3-1-0	4	-	4	ES	Data Structures and Algorithms	3-0-2	5	-	4
IE	Signals and Systems	3-1-0	4	-	4	ES	Design and analysis of Experiments	3-0-0	3	-	3
IE	Introduction to Machine Learning	3-0-2	5	-	4	ES	Modelling and Simulation	3-0-2	5	-	4
NE	Intro. To Profession	0-0-2	2	1		IH	Humanities I	3-0-0	3	-	3
	Total	**16-3-4**	**23**	**1**	**20**		**Total**	**17-0-6**	**23**	**-**	**20**
										7	**73**
										7	**-**
										7	**73**

Note: 2020 Batch B.Tech. students will do *Data Structures and Algorithms* in Semester III and *Thermodynamics* in Semester IV.

Table 3 The programme structure of 3rd and 4th years of B.Tech. in Engineering Science

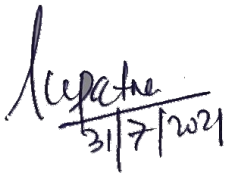
Cat	Course	LTP	CH	NC	GC	Cat	Course	LTP	CH	NC	GC
First two years same as Table 2 above											
	V Semester						**VI Semester**				
ES/OE	Engineering Science/Open Electives				17	ES/OE	Engineering Science/Open Electives				20
IH	Humanities II	3-0-0	3	-	3						
NH	Professional Ethics I	0-1-0		1	-	NH	Professional Ethics II	0-1-0		1	-
	Total			**1**	**20**		**Total**			**1**	**20**
	VII Semester						**VIII Semester**				
ES/OE	Engineering Science/Open Electives				17	ES/OE	Engineering Science/Open Electives				15
IS	Environmental Science	2-0-0	2	-	2	IH	Humanities III	3-0-0	3	-	3
	Total			**-**	**19**		**Total**			**-**	**18**
										9	**150**
										6	**-**
										15	**165**

Table 4 The programme structure of 3rd and 4th years of B.Tech. in X + Engineering Science

Cat	***Course***	***LTP***	***CH***	***NC***	***GC***	***Cat***	***Course***	***LTP***	***CH***	***NC***	***GC***
First two years same as Table 2 above											
	V Semester						**VI Semester**				
PC	**Programme Compulsory of X**				20	PC	**Programme Compulsory of X**				20
IH	Humanities II	3-0-0	3	-	3	PE / ES	Programme/Engineering Science Electives	3-0-0	3	-	3
NH	Professional Ethics I	0-1-0		1	-	NH	Professional Ethics II	0-1-0		1	-
	Total			**1**	**23**		**Total**	**15-2-8**		**1**	**23**
	VII Semester						**VIII Semester**				
PP	B. Tech. Project	0-0-6	6	-	3	PC	**Programme Compulsory of X**				6
PC	**Programme Compulsory of X**				4	PE / ES	Programme /Engineering Science Electives	17-0-0	17	-	11
PE / OE	Programme/Engineering Science Electives	9-0-0	9	-	12	IH	Humanities III	3-0-0	3	-	3
IS	Environmental Science	2-0-0	2	-	2						
	Total	**15-0-6**		**-**	**21**		**Total**	**20-0-0**	**20**	**-**	**20**
Total of graded and Non-Graded Credit										**9**	**160**
Non-Graded Design Credits										**6**	**-**
Grand Total										**15**	**175**

Note: Each Department will share positioning of Programme compulsory of X for due approval.

This issues with the approval of the Competent Authority.



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